

Parle Tilak Vidyalaya Association's
MULUND COLLEGE OF COMMERCE (AUTONOMOUS)
T.Y. B.Sc. Information Technology
Semester V

Course: Artificial Intelligence
Duration: 40 min.

Date: 8th October, 2024
Max Marks: 20

Answer any four of the following:

20

1. What is Artificial Intelligence? Explain its four approaches in detail.
2. Formulate 8-puzzle problem for the following scenario and also mention the steps and path cost required to reach goal state.

1	2	3
4	8	-----
7	6	5

Initial State

1	2	3
4	5	6
7	8	-----

Goal State

3. Define Alpha-Beta Pruning, explain its role in solving minimax algorithm problems, and provide an example with a game search tree.
4. Define Resolution, summarize the steps for Resolution, and convert FOL into Conjunctive Normal Form (CNF) using provided statements.
 - i) Convert to FOL.
 - ii) Convert FOL to CNF.
 - iii) Prove that "Is Someone Smiling?" Using Resolution.
 - iv) Draw Resolution tree.

Statements:-

- All people who are graduating are happy.
 - All happy people smile.
 - Someone is Graduating.
5. Define Semantic network, summarize it, and illustrate it with a suitable example.
 6. Differentiate between propositional and predicate logic.